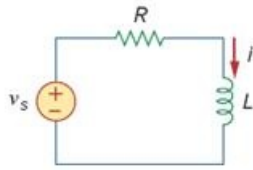


Problem

Using Fig. 18.40, design a problem to help other students better understand using Fourier transforms to do circuit analysis.

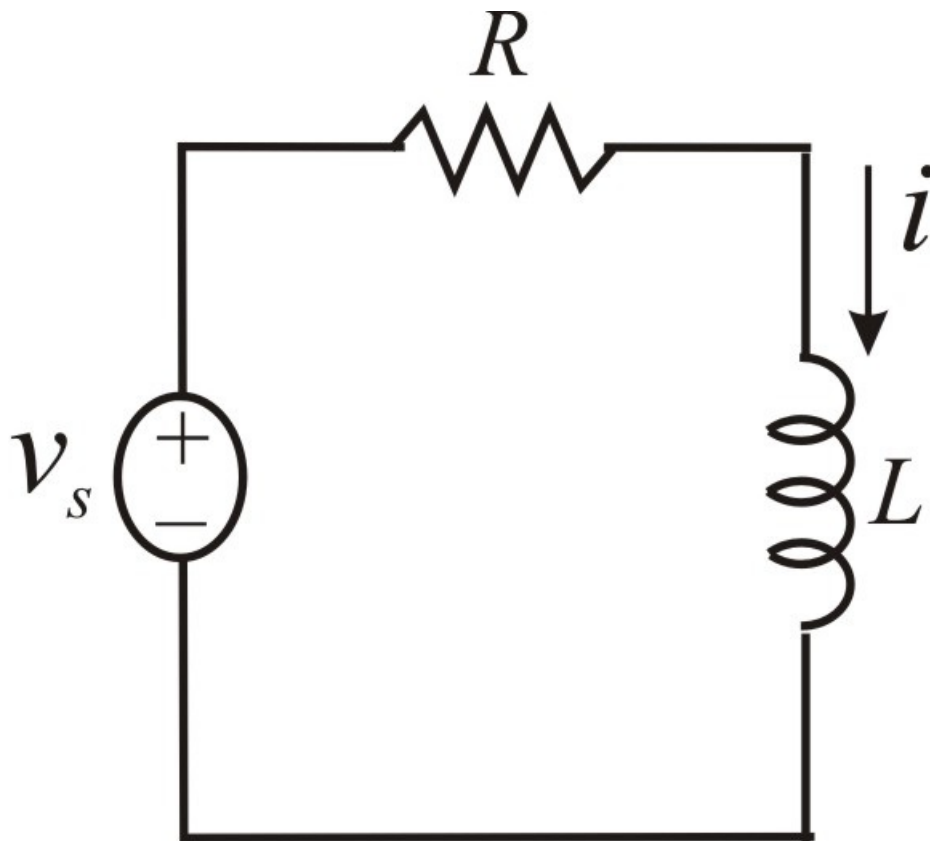
Figure 18.40



Step-by-step solution

Step 1 of 5

The given circuit is



Step 2 of 5

In frequency domain,

$$R \rightarrow R$$

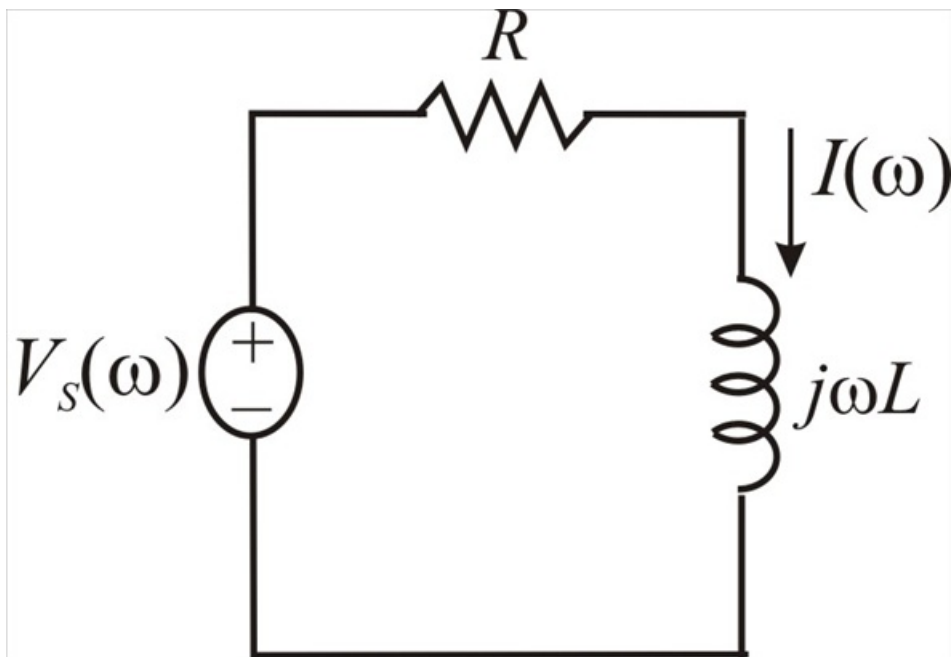
$$L \rightarrow j\omega L$$

$$v_s \rightarrow V_s(\omega)$$

$$i \rightarrow I(\omega)$$

Step 3 of 5

The circuit in frequency domain is



Step 4 of 5

The current flowing through the circuit is

$$I(\omega) = \frac{V_s(\omega)}{R + j\omega L}$$

$$I(\omega) = \frac{V_s(\omega)}{L \left(\frac{R}{L} + j\omega \right)}$$

Step 5 of 5

Taking inverse Fourier transform on both sides, we get

$$\mathcal{F}^{-1}\{I(\omega)\} = \mathcal{F}^{-1}\left\{ \frac{V_s(\omega)}{L \left(\frac{R}{L} + j\omega \right)} \right\}$$

$$\therefore i(t) = \frac{1}{L} \mathcal{F}^{-1}\left\{ \frac{V_s(\omega)}{\left(\frac{R}{L} + j\omega \right)} \right\} \text{ A}$$